Department for Education

Information Model Assurance Guide

Shared resources

May 2026

Contents

Summary

The Department for Education (DfE) Information Model Assurance Guide sets out an exemplary approach for performing automated compliance checks of 3D digital models using a set of preconfigured IDS files aligned to DfE's information requirements.

Review date

This document shall normally be reviewed at 12-month intervals however, DfE reserve the right to update it at any time.

Who this publication is for?

This guidance is for:

- technical professionals involved in the design and construction of school and college premises

Acknowledgements

This shared resource represents the collaborative efforts of the following organisations:

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- BAM
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- Bowmer + Kirkland
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- Mace
- Majenta
- Morta
- Morgan Sindall
- IIMBE
- Skanska
- Solibri
- Turner and Townsend
- Willmott Dixon
- xBIM
- yBIM

Document history

This table lists the key changes in each update

Revision code	Status code	Date	Amendment
P01	S3	2025-08-16	Initial version for review and comment
P02	S3	2026-05-20	Refactor based on IDS for review and comment

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1 Introduction

Projects delivered for the Department for Education (DfE) are required to meet a set of structured information requirements, many of which depend on the consistent and correct use of Industry Foundation Classes (IFC). Checking whether a 3D digital model meets these requirements forms a vital part of quality assurance and handover readiness.

Models shall be checked at the requisite project stages to confirm that the data present is correct, structured and formatted in accordance with the DfE's information requirements. Whilst it is not mandatory to use this guide or its accompanying resources to carry out these checks they do provide users with a consistent approach.

This guide refers to Information Delivery Specification (IDS), a free and open buildingSMART standard for defining and checking information requirements within IFC-SPF models. The purpose of this guide is not to promote specific platforms/software but to enhance the understanding and quality of IFC-SPF outputs for DfE funded projects and to help raise the standard of information delivery across the supply chain.

2 Overview

2.1 Purpose of the resources

This guide presents an approach to performing automated validation using a set of IDS files designed to assess an IFC-SPF (model) compliance with DfE's information requirements. The resources have been created with open standards (IDS) supporting interoperability and future expansion.

The files enable automated checking of models against the DfE's alphanumerical information requirements including naming conventions, classifications and required values.

Each IDS can be tailored to suit the needs of a specific project and to help verify whether expected values and formatting have been applied correctly. This allows project teams to check their models before submission and ensure they meet the requirements set by DfE.

2.2 Who should use this guide

This guide is intended for individuals and organisations responsible for producing and assuring IFC-SPF outputs in compliance with the DfE's information requirements. The resources are particularly aimed at those who do not yet have an established and consistent method for checking the alphanumerical information within their models prior to submission and provide an early warning system for potential non-compliance.

A basic working knowledge of IFC and model authoring processes is required. Where users are unfamiliar with these aspects, training is recommended.

3 Scope of the checks

3.1 Coverage

The IDS files provided with this guide have been developed to align with the delivery stages set out in DfE's Exchange Information Requirements (EIR) which correspond with the information exchanges across the RIBA Plan of Work stages.

The IDS files have been organised per stage and aligned to meet the specific requirements, with each set packaged as a separate file to ensure clarity and relevance for its intended phase of delivery. The following stage specific resources are available:

- Stage 3 – Spatial Coordination
- Stage 4 – Technical Design
- Stage 5 – Manufacturing and Construction

Due to the number of specifications, additional IDS files are also available, including a Core-only combined variant (covering shall requirements), a Nomenclature-only combined variant (covering naming and classification), as well as individual IDS files and entity grouped files.

3.2 Project specific requirements

Each project delivered under the DfE Construction Framework will include a project specific Detailed Exchange Information Requirements as part of its contract documentation. The requirements are standardised; however, individual projects may include amendments or clarifications to reflect specific appointments. These adjustments are agreed between the Contractor and DfE.

Where amendments exist (e.g., a change in Object Occurrence nomenclature) that would impact the checks described in this guide, adjustments and/or supplements to the IDS files may be required.

3.3 Limitations

These IDS files have been developed in collaboration with industry experts and are based on a shared interpretation and consensus of the requirements. There are, however, several important exclusions and clarifications regarding these resources:

- **IDS files must be tailored to the project:** The IDS files provided are generic and not preconfigured with project specific values. To ensure accurate results, users must replace all tokens with values from their Project's Information Standard (See Section 6)
- **IDS files are not a turnkey solution:** Users must integrate and test the IDS files within their own project environments to ensure outputs are valid, relevant and meaningful in the context of their workflow and data structure
- **IDS files do not check geometrical compliance:** The IDS files verify the presence, accuracy, formatting and consistency of structured alphanumerical data within IFC-SPF outputs, they do not verify geometrical compliance
- **IDS files are not intended for federated coordination:** Contractors and project teams may still find the resources useful for understanding how certain requirements have been interpreted and for identifying common areas of non-compliance

4 Getting started

4.1 What are the shared resources?

Each release will include the following:

Information Model Assurance Files: A set of preconfigured IDS files aligned with the DfE's Detailed Exchange Information Requirements (DEIR) and structured according to the RIBA Plan of Work stage[1]. These include[2]:

- ER-DFE-XX-XX-L-X-0030-Information Model Assurance Stage 3
- ER-DFE-XX-XX-L-X-0040-Information Model Assurance Stage 4
- ER-DFE-XX-XX-L-X-0050-Information Model Assurance Stage 5

Information Model Assurance Guide: Guidance on setting up, configuring and using the IDS files.

Token Replacer Tool: A browser-based HTML tool that allows users to load the IDS template files, enter project specific values, and download configured IDS files ready for use.

Read Me: Specifying any known compatibility issues, change logs and configurations relevant to the current version of the IDS files.

4.2 Where to download

The latest version of the resources referenced in this guide can be downloaded from here or from the relevant Microsoft Teams central resources area.

4.3 Compatibility

The IDS files conform to the IDS 1.0 specification published by buildingSMART International. It is important to ensure the correct version of IDS in your chosen tool is supported, as certain features required by the IDS files may not function properly in legacy/unpublished versions of the IDS specification.

[1] Each stage-specific ruleset contains only the checks relevant to that stage. DfE’s information requirements build progressively.

[2] Additional IDS files (Core, Nomenclature and Classification, individual and entity grouped) are also available.

5 Understanding the IDS files

5.1 What is IDS?

The Information Delivery Specification (IDS) is a free, open standard published by buildingSMART International for defining and checking information requirements within IFC models. An IDS file is an XML document that describes:

- **Applicability:** which IFC entities a check applies to (e.g. IfcProject, IfcSpace)
- **Requirements:** what properties, values or classifications those entities must have

IDS files are human-readable and can be opened in any text editor but are typically loaded into a compatible checking tool for automated validation against an IFC-SPF model. Further information about the IDS standard can be found [here].

An example of a simple IDS specification is shown below. This checks that every **IfcProject** entity has a **Name** attribute populated with the **project's** correct name:

```
<ids:specification name="Project Shall Have Name Matching The Projects Information
Standard">
 <ids:applicability minOccurs="1" maxOccurs="unbounded">
  <ids:entity>
   <ids:name>
    <xs:enumeration value="IFCPROJECT" />
   </ids:name>
  </ids:entity>
 </ids:applicability>
 <ids:requirements>
  <ids:attribute cardinality="required">
   <ids:name>
    <ids:simpleValue>Name</ids:simpleValue>
   </ids:name>
   <ids:value>
    <ids:simpleValue>{{IfcProjectName}}</ids:simpleValue>
   </ids:value>
  </ids:attribute>
 </ids:requirements>
</ids:specification>
```

Note: The {{IfcProjectName}} placeholder. This is a **token** that must be replaced with the actual project name before the IDS file can be used for checking

5.2 How the files are organised

The IDS files are organised into three tiers to suit different checking approaches:

- **Combined files:** Single files containing multiple rules
  - **Full Combined:** All checks for a given RIBA stage in a single file
  - **Core Only:** Shall checks for a given RIBA stage in a single file
  - **Nomenclature and Classification Only:** Naming conventions and classification checks for a given RIBA stage in a single file
- **Grouped files:** One file per entity group within each stage (e.g. Project, Site, Building, Building Storey, Space, Zone, Object Type). Useful when checking a specific area of the model
- **Individual files:** One file per specification rule

5.3 Token placeholders

The IDS template files contain t**oken placeholders**, values enclosed in double curly braces ({{ }}), that represent project-specific information which differs from project to project. These tokens must be replaced with actual values from your project's Information Standard before the IDS files can be used for checking.

If tokens are not replaced, the IDS checker will compare model data against the literal token string (e.g. {{IfcProjectName}}) which will always fail[3].

The full list of tokens used across the IDS files is shown below:

Token	Description
{{IfcProjectName}}	Project Name
{{IfcProjectDescription}}	Project Description
{{IfcProjectPhase}}	The current RIBA Stage
{{IfcSiteName}}	Site Name
{{IfcSiteDescription}}	Site Description
{{IfcBuildingName}}	Building Name
{{IfcBuildingDescription}}	Building Description
{{IfcBuildingClassificationReference}}	Building Uniclass Classification
{{BuildingBlockConstructionType}}	Block Construction Type

[3] If the case, spelling or punctuation does not match what is in the IFC model the rule will return a failure even if the model is otherwise correct.

Token	Description
{{IfcBuildingUPRN}}	Building Unique Property Reference Number (UPRN)
{{IfcBuildingStorey.Level 00.Elevation}}	Level 00 Building Storey Elevation Value
{{IfcBuildingStorey.Level 01.Elevation}}	Level 01 Building Storey Elevation Value
{{IfcBuildingStorey.Level 02.Elevation}}	Level 02 Building Storey Elevation Value
{{IfcBuildingStorey.Level 03.Elevation}}	Level 03 Building Storey Elevation Value
{{IfcBuildingStorey.Level 00.Height}}	Level 00 Building Storey Height Value
{{IfcBuildingStorey.Level 01.Height}}	Level 01 Building Storey Height Value
{{IfcBuildingStorey.Level 02.Height}}	Level 02 Building Storey Height Value
{{IfcBuildingStorey.Level 03.Height}}	Level 03 Building Storey Height Value

6 Configuring IDS files for your project

Before running the checks, you must replace all token placeholders with your project-specific values. Two methods are provided:

- Method A: Token replacer (Recommended)
- Method B: Manual find and replace

Both methods read the same token values and produce IDS files with tokens replaced by real project data.

6.1 Method A: Token replacer

The [Token Replacer]() is a browser-based tool provided as part of the shared resources. It requires no installation and runs entirely within a modern web browser (Chrome, Edge or Firefox).

6.1.1 Step 1: Open the tool

Clone the GitHub repo and navigate to the 'TokenReplacer/' folder within the resources and open TokenReplacer.html by double-clicking the file, or by dragging it into an open browser window.

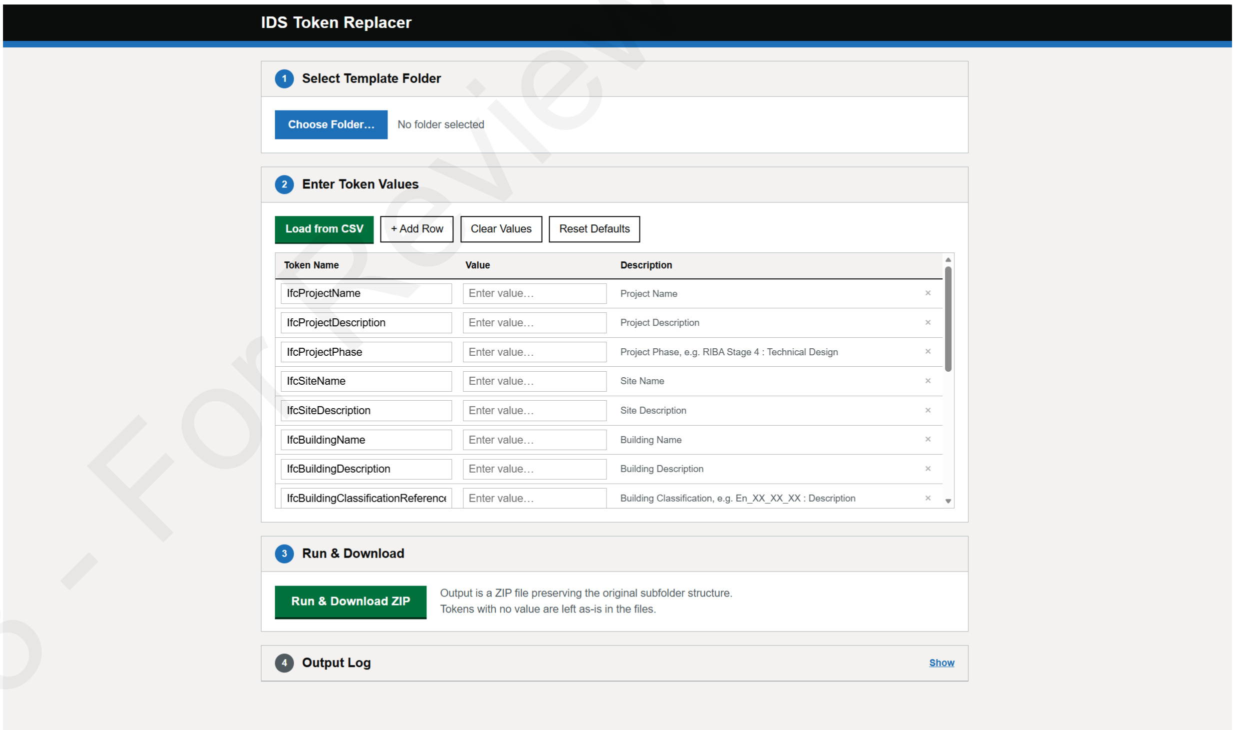

Figure 1: Token Replacer Interface

6.1.2 Step 2: Load the IDS template folder

Click **Choose folder** (Step 1 of the tool) and navigate to the IDS folder you wish to configure. Select the folder and the tool will scan it recursively for all .ids files containing token placeholders and display the tokens it has found.

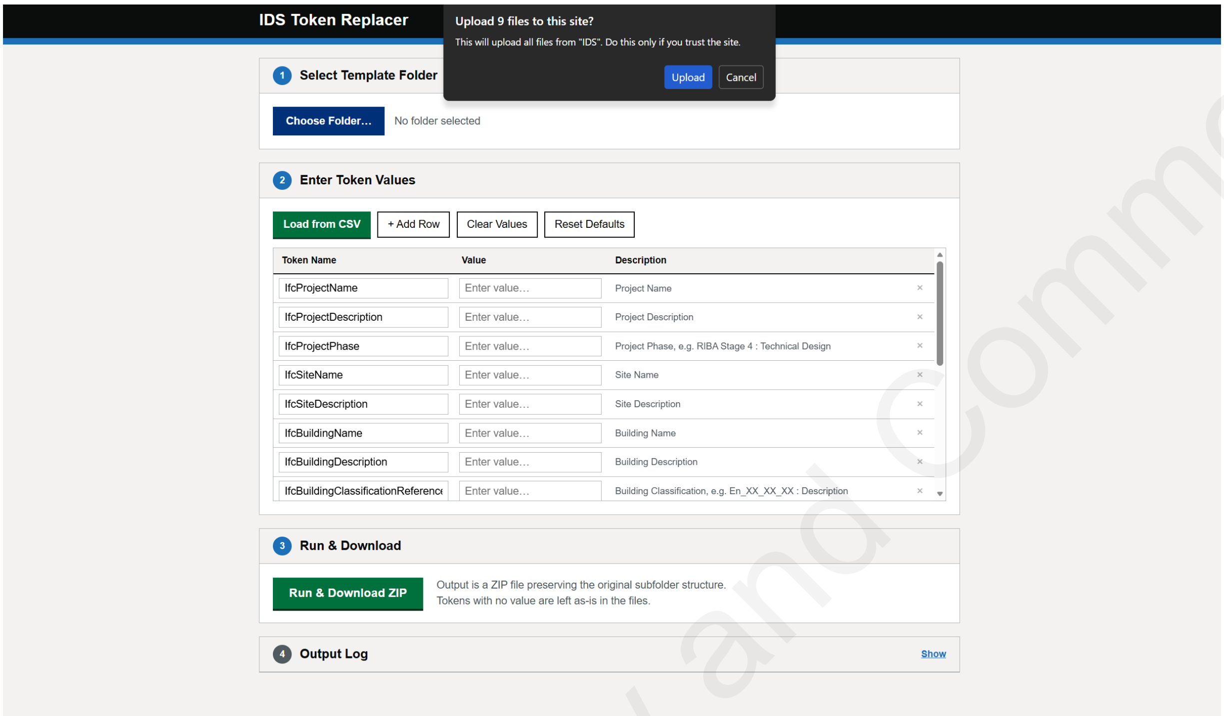

Figure 2: Confirmation Pop Up

6.1.3 Step 3: Enter project values

The tool displays a table of all unique tokens found across the loaded IDS files. For each token:

- Confirm or edit the Token name (the {{...}} identifier)
- Enter the Value - the project-specific replacement text

Refer to your Project's Information Standard for the correct values. The description column provides guidance on what each token represents.

Note: This can be completed more quickly using the load from CSV (refer to example.csv)

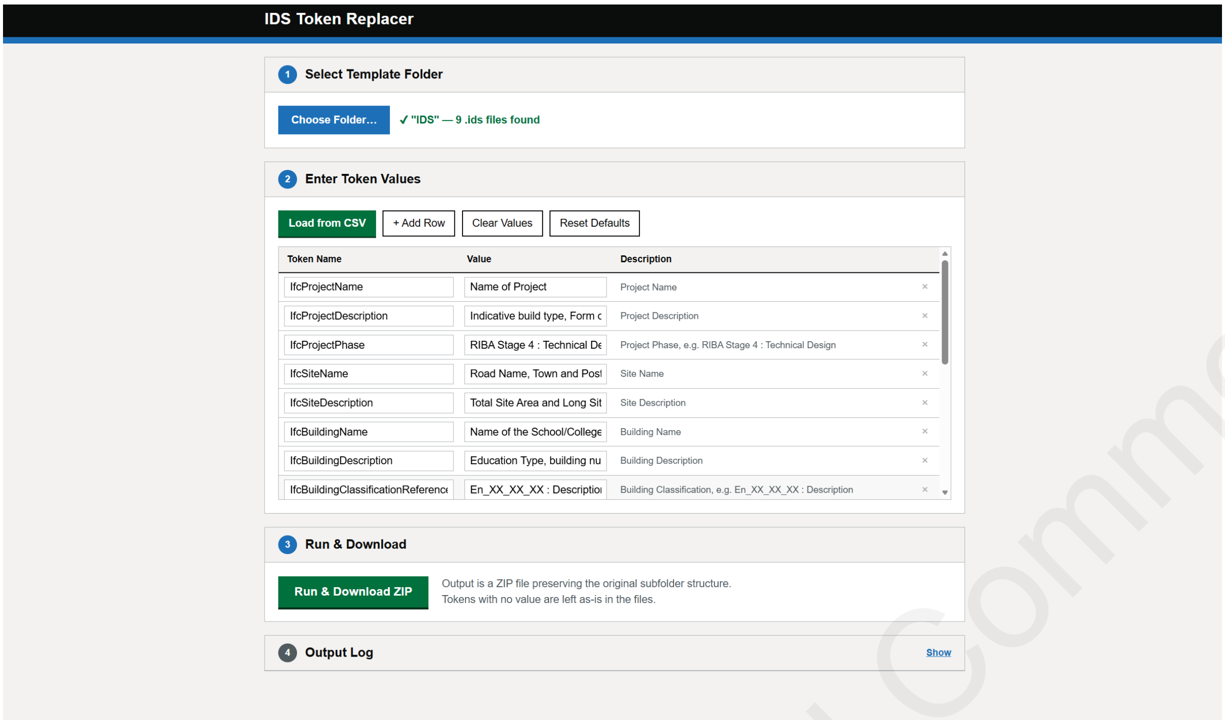

Figure 3: Project-specific Value Entered

6.1.4 Step 4: Generate project-specific IDS files

Once all tokens have been given values, click **Run & Download ZIP**. The tool will:

- Process all loaded IDS template files
- Replace every token placeholder with the corresponding value
- Download a ZIP file containing the configured IDS files, preserving the original folder structure

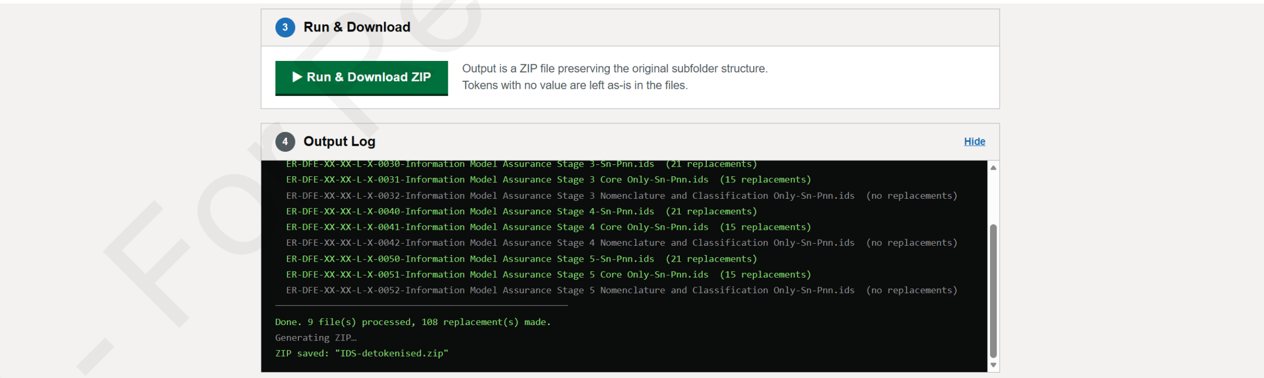

Figure 4: Example of successful output

The downloaded ZIP file contains your project-ready IDS files. Extract these to a convenient location before proceeding to Section 7.

6.2 Method B: Manual find and replace

If you prefer not to use the Token Replacer tool, you can replace tokens directly in each IDS file using the find and replace function built into any text editor (such as Notepad++, Visual Studio Code or Notepad).

Note: This method requires you to repeat the process for **every** IDS file you intend to use. Method A is recommended if you need to configure multiple files.

6.2.1 Step 1: Open the IDS file in a text editor

Right-click the .ids file you wish to configure and open it with a text editor. IDS files are plain XML and can be read and edited in any text editor.

6.2.2 Step 2: Open find and replace

Open the find and replace dialog in your text editor.

6.2.3 Step 3: Replace each token

For each token listed in the table in Section 5.3:

- Enter the token (e.g. {{IfcProjectName}}) in the Find field
- Enter your project-specific value in the Replace field
- Click Replace All
- Repeat for each remaining token

6.2.4 Step 4: Save the file

Save the file once all tokens have been replaced. It is recommended to save the configured file under a new name or into a separate folder so that the original template is preserved for future use. Repeat Steps 1 to 4 for each IDS file you intend to use, then proceed to Section 7.

7 Running the checks

Once you have configured IDS files with your project-specific values (using either method in Section 6), you can run the checks against your IFC model.

Refer to buildingSMART for a list of software vendors supporting IDS. The broad principles are the same across tools, load the IFC-SPF, load the configured IDS file(s) and run the check however, refer to each tool's own documentation for specific instructions. The steps below use Bonsai, an Open Source Blender Addon, as an example, as it is a widely available free tool with a graphical interface.

The guide focuses on standard layouts and default panels. Depending on how your workspace is arranged (e.g., which windows are docked or undocked, how panels are resized or whether you are using single or multiple monitors), your screen may appear slightly differently to the screenshots in this guide.

7.1 Which IDS file to use:

- Use the **Core Only** file for a quick baseline check that required data is present
- Use the **Nomenclature and Classification Only** file to check naming and classification conventions
- Use the **full combined** file for a comprehensive single-pass check
- Use **Grouped** or **Individual** files when investigating a specific entity or rule

7.2 Bonsai Example

7.2.1 Step 1: Open your IFC model

Launch Bonsai and open the IFC-SPF model you wish to check.

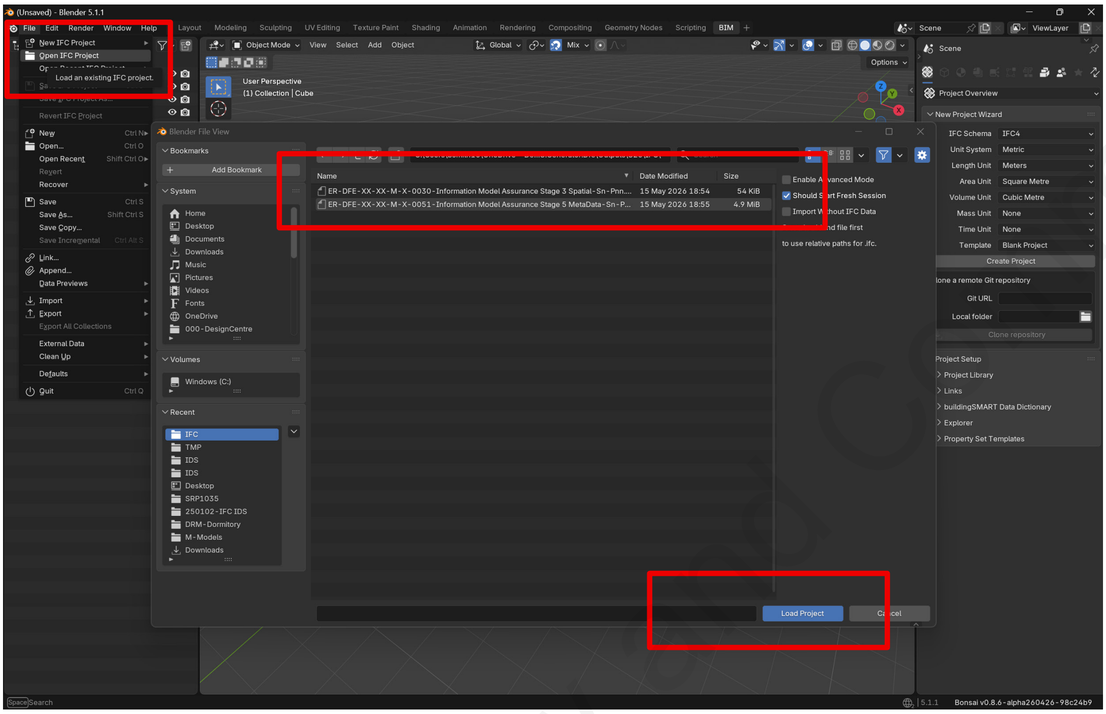

Figure 5: Load IFC File

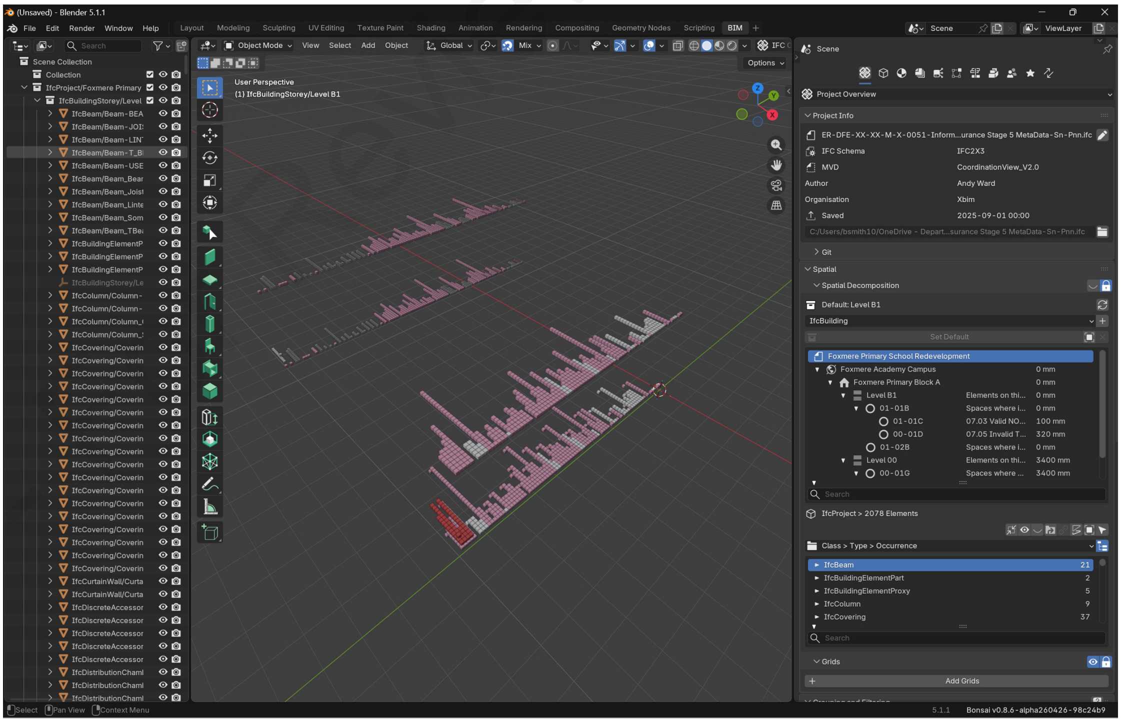

Figure 6: IFC-SPF Loaded

7.2.2 Step 2: Load the IDS file

Navigate to the ‘Quality and Coordination’ panel within Bonsai and expand IFC Tester. Select ‘Load from Memory’, ‘Generate HTML Report’ and then press the folder icon against ‘IDS File(s)’. This will expand a pop up file browser for you to select the relevant IDS file(s) you want to run against the model.

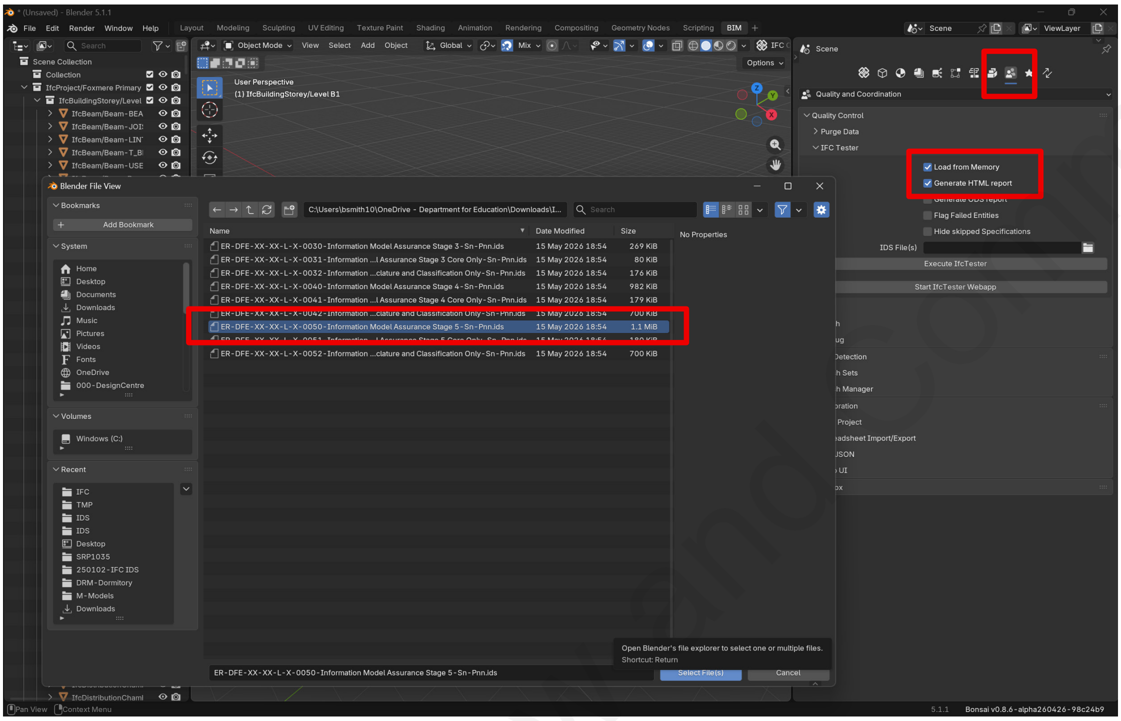

Figure 7: Load IDS File(s)

7.2.3 Step 3: Run the check

Press the ‘Execute IfcTester’ button within the current panel. Bonsai will process the model against all specifications in the loaded IDS file.

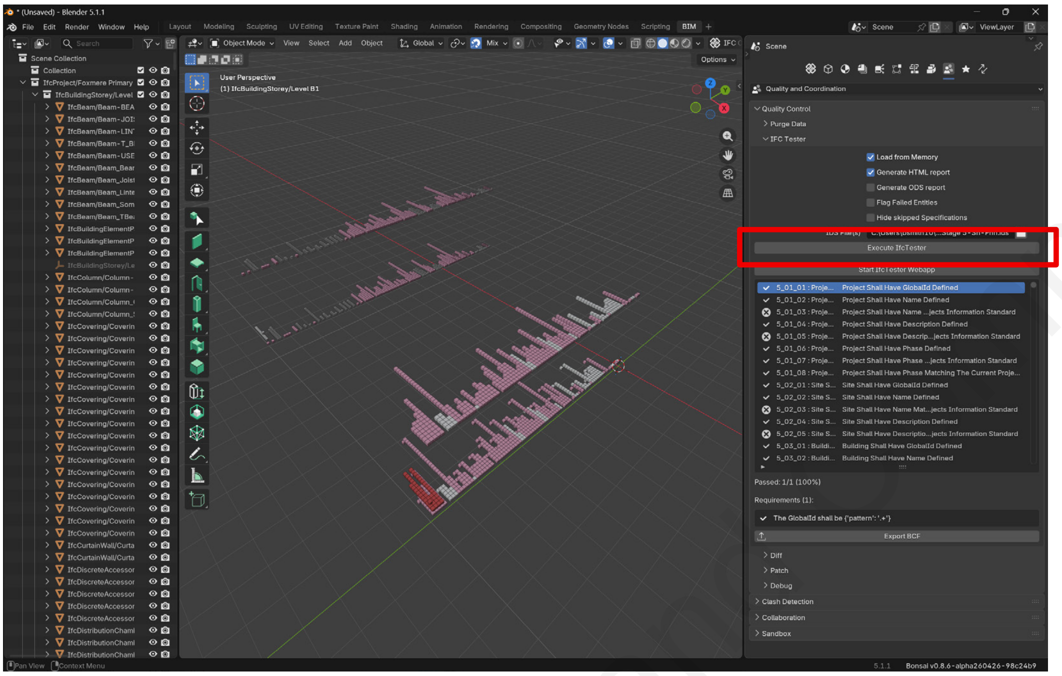

Figure 8: Run the check

Note: Subject to model size and number of IDS files selected this may take a minute to process.

7.2.4 Step 4: Review the results

Once complete, the results will be displayed within the tool. A new sub-section will appear in the panel, enabling you to identify which attributes, properties, or classification information in the model are non-compliant. Proceed to Section 8 for guidance on interpreting the output.

8 Understanding the results

8.1 Interface

A new HTML window with the results is generated laying out each pass/fail/n/a in detail. Each specification in the IDS file will show one of the following outcomes:

Indicator	Meaning
Pass (green)	The specification has passed. All applicable model objects meet the requirement
Fail (red)	The specification has failed. One or more applicable objects do not meet the requirement
n/a (grey)	No objects in the model match the applicability criteria. This is expected for certain rules depending on model content

All failures should be reviewed individually to understand the cause and determine what correction is required in the model.

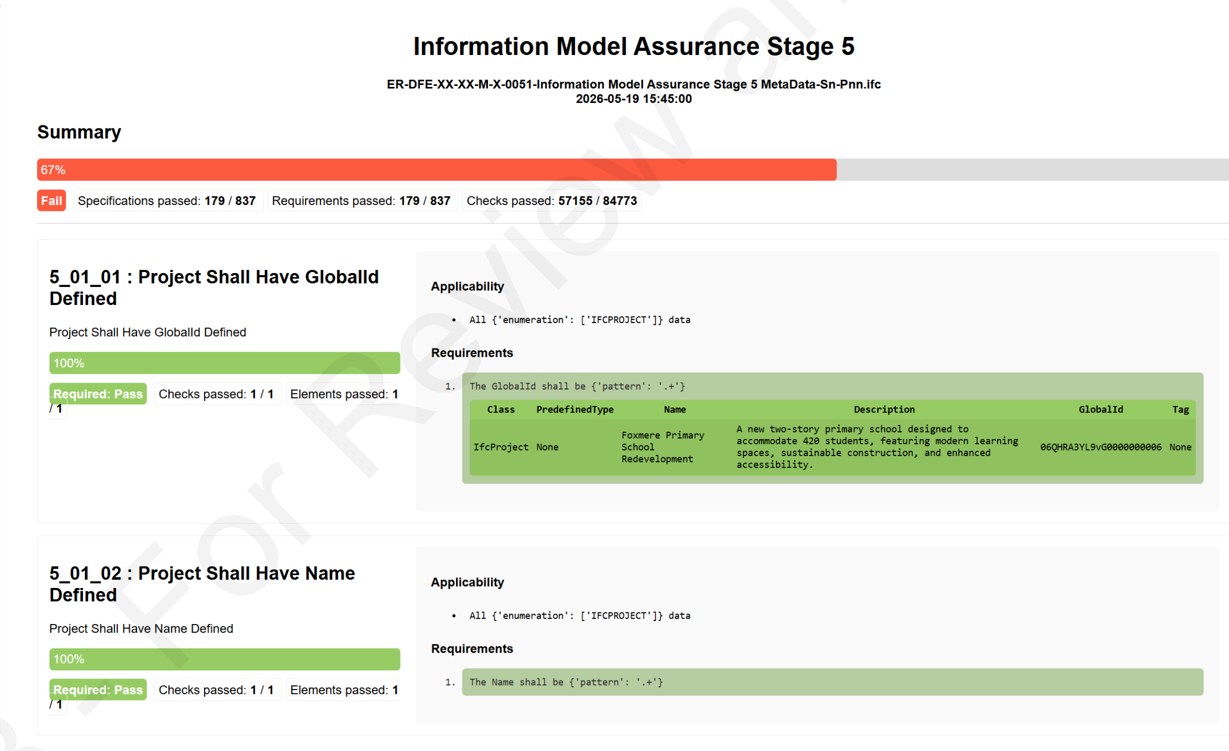

Figure 9: Result Output

8.2 Investigating a failure

When a specification fails, clicking on it will typically display:

- **The object that failed:** The specific IFC entity (e.g. IfcBuilding, IfcSpace) that did not meet the requirement
- **The expected value:** What the IDS specification required
- **The value found in the model:** What was actually present in the IFC

By comparing the found and expected values, you can determine what correction needs to be made in the authoring tool.

8.3 Exporting results

Most IDS compatible platforms provide options to export results as reports (HTML, JSON, BCF or CSV formats). Refer to your chosen tool's documentation for specific export instructions. Reports can be used to communicate non-compliance items back to the model author or to maintain an audit record.

9 Frequently asked questions

Below are some common questions and issues users may encounter when using this guide and the accompanying resources:

My layout looks different from what's shown in this guide

The screenshots in this guide are based on a specific version of Blender (5.1). Layouts may vary between versions and can be customised by users. The layout shown is the ‘standard’ interface loaded following installation of Bonsai.

An IDS check shows as passed but I know the value is wrong

- Check that all tokens have been replaced correctly, particularly that values match the Project's Information Standard exactly
- Confirm that the correct IDS file has been loaded for the corresponding RIBA stage
- Ensure you are checking the correct discipline model
- If everything appears correct and the issue persists, use Section 10 Feedback to report it

An IDS check is flagging a failure but I believe my model is correct

- Confirm that token values in the IDS file exactly match what is in the model, including capitalisation, spacing and punctuation
- Check whether the failure is a false positive caused by a subtle formatting difference (e.g. a trailing space, or Stage3 versus Stage 3)
- If you are confident the model is correct, report the issue via Section 10 Feedback to help improve the specification logic

Tokens are still showing in the expected values column

This means the IDS file has not been de-tokenised. Either the token replacement process was not completed, or it was run against the wrong folder. Return to Section 6 and repeat the process, ensuring you select the correct IDS template folder.

The check is not finding any objects to test (all results are "not applicable")

- Confirm that the IFC model has been exported correctly and contains the expected entities (e.g. IfcProject, IfcSite, IfcBuilding, IfcBuildingStorey, IfcSpace)
- Check that the IFC schema version of the model matches the version targeted by the IDS file (the DfE IDS files target IFC2X3)
- Some rules only apply to specific disciplines. Confirm you are running the correct IDS file against the appropriate model

My project has slightly different naming conventions to those in the Project's Information Standard

The IDS files are designed to check compliance against the DfE's information requirements as set out in the Project's Information Standard and Detailed Exchange Information Requirements. If your project uses different formats, agreed in advance with DfE, update the IDS specification accordingly.

Where should I go if I am stuck or need support?

Start by reviewing the relevant sections of this guide. If your issue is not covered by this guide or persists, use the feedback mechanism in [Section 10]

10 Feedback

Feedback can be submitted where errors are found or where clarification may be required via GitHub Issues, DfE's feedback form or email. Future updates to improve the resources will consider this feedback.

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